ISMÉTLÉS

A term. ábr. tér

S_1, S_2 A feltétlen prog.

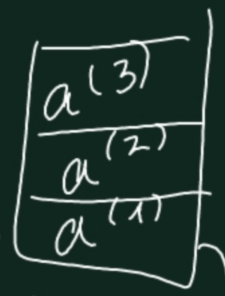
$S_1 \subseteq S_2$

$\boxed{D_{P(S_1)} \subseteq D_{P(S_2)}?}$ → ál.ban nem igaz

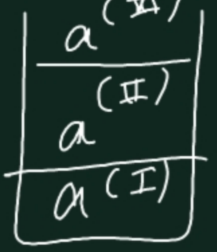
[ELLENPÉLDA]

...

$a \in D_{P(S_1)}$ →



$D_{P(S_1)}$



$D_{P(S_2)}$

$a \in D_{P(S_1)}$. Ekkor $S_1(a) \subseteq \bar{A}^*$. Viszont $S_1 \subseteq S_2$, tehát lehet olyan $b \in S_2(a)$, hogy $b \notin S_1(a)$, és az is lehet, hogy $b \notin \bar{A}^*$.

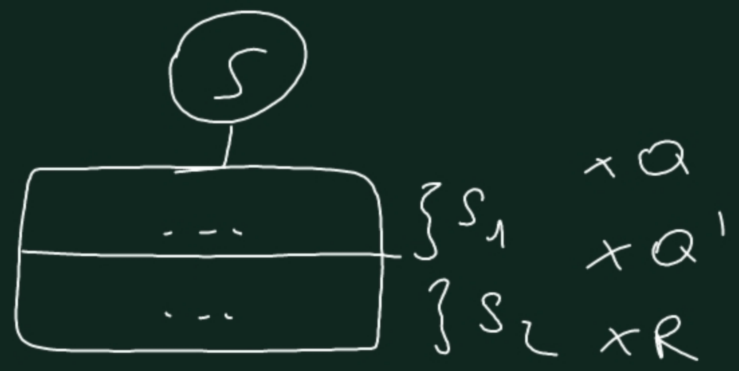
Tehát nem garantált, hogy ha $a \in D_{P(S_1)}$, akkor $a \in D_{P(S_2)}$.

$$\bigotimes_4 \left(\langle 1,2,3,\underline{1} \rangle, \langle \underline{1},2,3,\underline{1} \rangle, \langle \underline{1},2,3,\underline{1} \rangle, \langle \underline{1},2,3,1 \rangle \right)$$

$$1\ 2\ 3\ \underline{1}\ \cancel{1}\ 2\ 3\ \underline{1}\ \cancel{1}\ 2\ 3\ \underline{1}\ \cancel{1}\ 2\ 3\ 1$$

$$\bigotimes_4 \left(\langle \underline{1} \rangle, \langle \underline{1} \rangle, \langle \underline{1} \rangle, \langle \underline{1},2,3,1 \rangle \right)$$

$$\underline{1}\ \underline{1}\ \underline{1}\ \cancel{1}\ 2\ 3\ 1$$

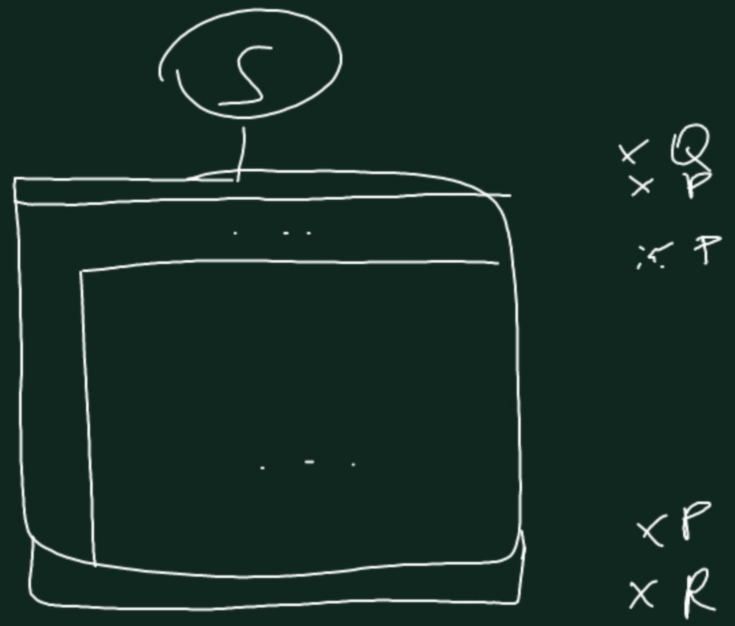


$Q \Rightarrow lf(S, R)$

helyett

① $Q \Rightarrow lf(S_1, Q')$

② $Q' \Rightarrow lf(S_2, R)$



$Q \Rightarrow lf(S, R)$

helyett

① $Q \Rightarrow P$

② $P \Rightarrow \Pi \vee \neg \Pi$

③ $P \wedge \neg \Pi \Rightarrow R$

④ $P \wedge \Pi \Rightarrow t > 0$

⑤ $P \wedge \Pi \wedge t = t_0 \Rightarrow$

$lf(S_0, P \wedge t < t_0)$

$(t_0 \in \mathbb{Z})$

$$F \begin{cases} A = (x : \mathbb{N}^+,\ l : \mathbb{L}) \\ B = (x' : \mathbb{N}^+) \\ Q = (x = x' \wedge x > 1) \\ R = (Q \wedge l = (\forall j \in [2..x-1] : j \nmid x)) \end{cases}$$

S A program állapottere $(x : \mathbb{N}^+,\ k : \mathbb{N}^+,\ l : \mathbb{L})$

S

$k, l := 2, igaz$

$k \nmid x$

$l := l \wedge k \nmid x$

$k := k + 1$

$\}S_1$

S_2

$\}\#$

S_0

$\}S_{0,\neq}$

$\}S_{0,\#}$

$\times Q_1$

$\times Q$

$\times P$

$\times Q''$

$\times P$

$\times R$

$$Q' = (Q \wedge k = 2 \wedge l = igaz)$$

$$Q'' = (Q \wedge l = (\forall j \in [2..k] : j \nmid x) \wedge k+1 \in [2..x] \wedge x - k = t_0)$$

$$P = (Q \wedge l = (\forall j \in [2..k-1] : j \nmid x) \wedge k \in [2..x])$$

$$t : x - k$$

Levezetési irányok

1// $Q \Rightarrow lf(S, R)$

helyett

(A) $Q \Rightarrow lf(S_1, Q')$

$Q' \Rightarrow lf(S_2, R)$

2// $Q' \Rightarrow lf(S_2, R)$

helyett

(B) $Q' \Rightarrow P$

(C) $P \Rightarrow T \vee \neg T$

(D) $P \wedge \neg T \Rightarrow R$

(E) $P \wedge T \Rightarrow t > 0$

$P \wedge T \wedge t = t_0 \Rightarrow lf(S_0, P \wedge t < t_0)$

3// $P \wedge T \wedge t = t_0 \Rightarrow lf(S_0, P \wedge t < t_0)$

helyett

(F) $P \wedge T \wedge t = t_0 \Rightarrow lf(S_{0,\neq}, Q'')$

(G) $Q'' \Rightarrow lf(S_{0,\neq}, P \wedge t < t_0)$

(A) $Q \Rightarrow wp(S_1, Q')$

$$wp(S_1, Q') = wp(\underbrace{k, l := 2, igaz}_{k \leftarrow 2,\ l \leftarrow igaz}, \underbrace{x = x' \wedge x > 1 \wedge k = 2 \wedge l = igaz}) =$$

$$= (x = x' \wedge x > 1 \wedge k = 2 \wedge l = igaz) =$$

$$= (x = x' \wedge x > 1 \wedge 2 = 2 \wedge igaz = igaz) = (x = x' \wedge x > 1) = Q$$

$Q \Rightarrow Q$ ✓

(B) $Q' \Longrightarrow P$

$x = x'$ ✓a

$x > 1$ ✓b ✓d

$k = 2$ ✓c ✓d

$l =$ igaz ✓c

$x = x'$ ✓a

$x > 1$ ✓b

$l = (\forall j \in [2..k-1] : j \nmid x)$ ✓c

$k \in [2..x]$ ✓d

(C) $k = 2$, $l =$ igaz esetén

$$\underbrace{l}_{\text{igaz}} = (\forall j \in [2..\underbrace{k-1}_{\underbrace{2-1}_{1}}] : j \nmid x)$$

$\emptyset$

igaz ✓

(d) $k = 2$, és $x > 1$ ha

$k \in [2..x]$

$2 \in [2..m]$, ahol $m \geq 2$

✓igaz

(c) $P \Rightarrow T \vee \neg T$

$x = x'$

$x > 1$

$\ell = (\forall j \in [2..k-1] : j \nmid x)$

$k \in [2..x]$

$k \neq x \vee k = x$ ✓a

(a) $k : \mathbb{N}^+$, $x : \mathbb{N}^+$, tehát $k \neq x \vee k = x$ ✓

(D) $P \wedge \neg T \Rightarrow R$

$x = x'$ ✓a

$x > 1$ ✓b

$\ell = (\forall j \in [2..k-1] : j \nmid x)$ ✓c

$k \in [2..x]$

$k = x$ ✓c

$x = x'$ ✓a

$x > 1$ ✓b

$\ell = (\forall j \in [2..x-1] : j \nmid x)$ ✓c

(c) $\ell = (\forall j \in [2..\underset{\overset{\parallel}{x}}{k}-1] : j \nmid x)$ és $k = x$ esetén

$\ell = (\forall j \in [2..x-1] : j \nmid x)$

(E) $P_n \Pi \Rightarrow t > 0$

$x = x'$

$x > 1$

$e = (\forall j \in [2..k-1] : j \nmid x)$

$k \in [2..x]$ ✓a

$k \neq x$ ✓a

$x - k > 0$ ✓a

(a) $k \in [2..x]$

$\hookrightarrow 2 \leq k \leq x$

(+) $\rightarrow k \neq x$

$2 \leq k < x$

$x - k > 0$

(F) $P \wedge \Pi \wedge t = t_0 \Longrightarrow wp(S_{0,I}, Q'')$

$$wp(S_{0,I}, Q'') = \Big(\big(x = x' \wedge x > 1 \wedge l = (\forall j \in [2..k] : j \nmid x) \wedge k+1 \in [2..x] \wedge x - k = t_0 \big)^{l \leftarrow l \wedge k \nmid x} \wedge k \neq 0 \Big) =$$

$$= \big(x = x' \wedge x > 1 \wedge (l \wedge k \nmid x) = (\forall j \in [2..k] : j \nmid x) \wedge k+1 \in [2..x] \wedge x - k = t_0 \wedge k \neq 0 \big)$$

(F) $P \wedge \pi \wedge t = t_0 \Rightarrow \mathrm{lf}(S_{0,I}, Q'')$

$x = x'$ ✓a

$x > 1$ ✓b

$l = (\forall j \in [2..k-1] : j \nmid x)$

$k \in [2..x]$ ✓d

$k \neq x$ ✓d

$x - k = t_0$ ✓e

$x = x'$ ✓a

$x > 1$ ✓b

$(l \wedge k \nmid x) = (\forall j \in [2..k] : j \nmid x)$ ✓c

$k + 1 \in [2..x]$ ✓d

$x - k = t_0$ ✓e

$k \neq 0$ ✓f

(f) $k : \mathbb{N}^+$, tehát $k \neq 0$

(c) $l = (\forall j \in [2..k-1] : j \nmid x)$

esetén $(l \wedge k \nmid x) =$

$= (\forall j \in [2..k-1] : j \nmid x \wedge k \nmid x) =$

$= (\forall j \in [2..k] : j \nmid x)$ ✓

(d) $k \neq x$, $k \in [2..x]$ esetén

$2 \leq k \leq x$

$2 \leq k < x$

$2 \leq k+1 \leq x$, tehát $k+1 \in [2..x]$

(6) $Q'' \Longrightarrow \mathrm{lf}(S_{0,\mathrm{II}}, P \wedge t < t_0)$

$x = x'$ ✓a

$x > 1$ ✓b

$l = (\forall j \in [2..k] : j \nmid x)$ ✓c

$k+1 \in [2..x]$ ✓d

$x - k = t_0$ ✓e

$x = x'$ ✓a

$x > 1$ ✓b

$l = (\forall j \in [2..k] : j \nmid x)$ ✓c

$k+1 \in [2..x]$ ✓d

$x - k - 1 < t_0$ ✓e

(e) $x - k = t_0$ esetén $x - k - 1 = t_0 - 1 < t_0$

$$\mathrm{lf}(S_{0,\mathrm{II}}, P \wedge t < t_0) = (x = x' \wedge x > 1 \wedge l = (\forall j \in [2..k-1] : j \nmid x) \wedge k \in [2..x] \wedge x - k < t_0)^{k := k+1} =$$

$$(x = x' \wedge x > 1 \wedge l = (\forall j \in [2..k] : j \nmid x) \wedge k+1 \in [2..x] \wedge x - k - 1 < t_0)$$

$\boxed{k := k+1}$